

Microprocessor and Computer Hardware

Rapid-Revision One-Liners

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Microprocessor and Computer Hardware — One-Liners

Microprocessor Fundamentals

1. The microprocessor performs **3** fundamental steps per instruction: **fetch**, **decode**, and **execute**.
2. The microprocessor has exactly **3** internal functional sections: the **ALU**, the **register array**, and the **control unit**.
3. The first microprocessor was the **Intel 4004**, released in **1971**, which was a **4-bit** device.
4. A microprocessor is fabricated using **VLSI** technology on a single **silicon** chip.
5. There are broadly **5** microprocessor generations by word size: **4-bit (4004)**, **8-bit (8085)**, **16-bit (8086)**, **32-bit (80386)**, and **64-bit (Pentium+)**.
6. The **8-bit** second-generation Intel processors are **3** in number: **8008**, **8080**, and **8085**.
7. A microprocessor differs from a microcontroller on **1** key point: the microprocessor holds **only the CPU**, while a microcontroller integrates **CPU + RAM + ROM + I/O + timer**.
8. The microprocessor usually follows **Von Neumann** architecture, whereas the microcontroller usually follows **Harvard** architecture.
9. Two common embedded microcontrollers are the **8051** and the **PIC/AVR** families.
10. The microprocessor performs **4** machine operations: **memory read/write**, **I/O read/write**, **arithmetic/logic**, and **branching**.
11. The defining word size of the **8085** is **8 bits**, meaning it processes **1 byte** at a time.
12. The complete set of commands a CPU understands is its **instruction set**, and the 8085 has **74** instruction types with **246** opcodes.

Bus Organisation

13. The system bus is made of **3** buses: the **address bus**, the **data bus**, and the **control bus**.
14. The **8085 address bus** is **16-bit** wide (A0–A15) and is **unidirectional**.
15. The **8085 data bus** is **8-bit** wide (D0–D7) and is **bidirectional**.
16. The address bus has **1** direction property — **unidirectional** — because addresses always flow **out** of the CPU.
17. With **16** address lines the 8085 can access $2^{16} = 64 \text{ KB} = 65,536$ memory locations.
18. Memory capacity is decided by **1** factor — the **address-bus width** (2^n) — not the data-bus width.
19. The control bus carries key signals such as **RD**, **WR**, **ALE**, and **IO/M**.
20. For I/O-mapped I/O the 8085 uses an **8-bit** port address giving **256** input ports and **256** output ports.

8085 / 8080A Architecture

21. The **8085** is an **8-bit** microprocessor introduced in **1976** using **NMOS** technology.
22. The 8085 chip contains about **6,500** transistors and is packaged in a **40-pin** DIP.
23. The 8085 needs just **1** power supply of **+5 V**, unlike the 8080A which needs **3** supplies.
24. The 8085A operates at a maximum clock frequency of **3 MHz**.
25. The 8080A requires **2** external support chips: the **8224** clock generator and the **8228** system controller.
26. The 8085 added exactly **2** instructions over the 8080A: **SIM** and **RIM**.
27. The 8085 has **5** hardware interrupts, whereas the 8080A has only **1** (INT).
28. The 8085 provides **2** serial-data pins absent in the 8080A: **SID** and **SOD**.
29. The 8085 control unit generates timing signals like **ALE**, **RD**, **WR**, and **IO/M** from its **timing and control** block.
30. The 8085 ALU works with **2** associated elements: the **accumulator** and the **temporary register**, and it sets the **flags**.

Registers

31. The 8085 has **6** general-purpose **8-bit** registers: **B**, **C**, **D**, **E**, **H**, and **L**.
32. The 8085 general-purpose registers form **3** pairs: **BC**, **DE**, and **HL**.
33. There is **1** special register pair used as a memory pointer — the **HL** pair, referenced as **M**.
34. The **accumulator (A)** is an **8-bit** register that holds **1** operand and stores the **result**.

35. The 8085 has **2** sixteen-bit registers: the **Program Counter (PC)** and the **Stack Pointer (SP)**.
36. The **Program Counter** is **16-bit** and holds the address of the **next** instruction to fetch.
37. The **Stack Pointer** is **16-bit** and points to the **top of the stack** in memory.
38. The 8085 has **2** programmer-inaccessible temporary registers: **W** and **Z**.
39. The **instruction register (IR)** stores **1** item — the current **opcode** — and is not accessible to the programmer.
40. Counting A and flags, the programmer-visible registers total: **6** GP + **1** accumulator + **1** flag + **2** sixteen-bit (PC, SP).

Flags

41. The 8085 flag register is **8-bit** wide but uses only **5** flags.
42. The **5** flags of the 8085 are: **Sign (S)**, **Zero (Z)**, **Auxiliary Carry (AC)**, **Parity (P)**, and **Carry (CY)**.
43. The flag bit positions are: **S = D7**, **Z = D6**, **AC = D4**, **P = D2**, **CY = D0**.
44. The flag register has **3** unused bits at positions **D5, D3, and D1**.
45. The **Sign flag (S)** is set when the result MSB (**D7**) is **1**, indicating a **negative** result.
46. The **Zero flag (Z)** is set when the result equals **00H** (zero).
47. The **Parity flag (P)** is set for an **even** number of 1s in the result.
48. The **Carry flag (CY)** is set on a carry/borrow out of bit **D7**.
49. The **Auxiliary Carry (AC)** records a carry out of bit **D3** and is used only by the **DAA** instruction.
50. Out of **5** flags, **4** can be tested by conditional jumps — **S, Z, P, CY** — while **AC** cannot.

Pins and Signals

51. The 8085 is housed in a **40-pin** dual in-line package.
52. The 8085 has **8** higher-order address lines **A8–A15** (pins **21–28**).
53. The 8085 has **8** multiplexed address/data lines **AD0–AD7** (pins **12–19**).
54. There is **1** signal used to latch the lower address — **ALE** (pin **30**) — which is high during **T1**.
55. The **RD** (pin 32) and **WR** (pin 31) control signals are both **active low**.
56. The **IO/M** signal selects between **2** spaces: **high = I/O** and **low = memory**.
57. The machine-cycle type is identified by **3** signals: **IO/M, S1, and S0**.
58. An opcode-fetch machine cycle is shown by **IO/M S1 S0 = 0 1 1**.
59. There are **2** DMA-handshake pins: **HOLD** (request, pin 39) and **HLDA** (acknowledge, pin 38).
60. The **READY** pin (pin 35) makes the CPU enter **wait** states when it is **low**.
61. Power pins are **2**: **Vcc = +5 V** (pin 40) and **Vss = ground** (pin 20).
62. Clock pins are **3**: **X1** (pin 1), **X2** (pin 2), and **CLK OUT** (pin 37).
63. The internal clock frequency equals the crystal frequency divided by **2**, so a **6 MHz** crystal gives **3 MHz**.
64. There are **2** serial pins: **SID** (pin 5, input) and **SOD** (pin 4, output).
65. There are **2** reset pins: **RESET IN** (pin 36, active low) and **RESET OUT** (pin 3).
66. Demultiplexing **AD0–AD7** requires **1** external latch — the **74LS373** — enabled by **ALE**.

Instruction Format and Classification

67. The 8085 has **3** instruction sizes: **1-byte**, **2-byte**, and **3-byte**.
68. A **1-byte** instruction holds **only the opcode**, e.g., **MOV A,B** or **CMA**.
69. A **2-byte** instruction holds opcode + **8-bit** data, e.g., **MVI A, 32H**.
70. A **3-byte** instruction holds opcode + **16-bit** address/data, e.g., **LDA 2050H** or **JMP 2000H**.
71. In a 3-byte instruction the order is: byte 2 = **lower** address and byte 3 = **higher** address (little-endian).
72. The 8085 instruction set is divided into **5** groups: **data transfer, arithmetic, logical, branching, and machine-control**.
73. Data-transfer instructions like **MOV, MVI, LDA, STA, LXI** affect **0** flags (they only copy).
74. Arithmetic instructions include **ADD, SUB, INR, DCR, DAD, DAA** and they update the flags.
75. Logical instructions include **ANA, ORA, XRA, CMP, RLC, RRC, RAL, RAR, CMA**.

- 76. Branching instructions include **JMP, CALL, RET, RST, and PCHL**.
- 77. Machine-control instructions are **6: NOP, HLT, DI, EI, SIM, RIM**.
- 78. The instruction that adjusts the accumulator after BCD addition is **1 — DAA** — and it uses the **AC** flag.

Addressing Modes

- 79. The 8085 supports **5** addressing modes: **immediate, register, direct, register-indirect, and implied**.
- 80. **Immediate** addressing supplies data within the instruction, e.g., **MVI A, 05H** and **LXI H, 2000H**.
- 81. **Register** addressing uses CPU registers as operands, e.g., **MOV A, B** and **ADD C**.
- 82. **Direct** addressing specifies a **16-bit** memory address, e.g., **LDA 2000H** and **STA 2000H**.
- 83. **Register-indirect** addressing uses a register pair to hold the address, e.g., **MOV A, M** and **LDAX B**.
- 84. **Implied** addressing names **no** operand explicitly, e.g., **CMA, RAL, STC, and CMC**.

Stacks and Subroutines

- 85. The stack follows **1** discipline — **LIFO** (Last In, First Out).
- 86. In the 8085 the stack grows toward **lower** memory addresses (downward).
- 87. The stack top is tracked by **1** register — the **16-bit Stack Pointer (SP)**.
- 88. **PUSH** decrements SP by **2** and stores the **high** byte first, then the **low** byte.
- 89. **POP** increments SP by **2** and restores the **low** byte first, then the **high** byte.
- 90. Stack operations work on **4** register pairs: **BC, DE, HL, and PSW**.
- 91. **PSW** combines **2** registers: the **accumulator** and the **flag register**.
- 92. A **CALL** instruction is **3** bytes and takes **18** T-states, pushing the return address onto the stack.
- 93. A **RET** instruction takes **10** T-states and pops the return address back into the **PC**.
- 94. Nesting of subroutines is managed by **1** structure — the **stack** — preserving return addresses in **LIFO** order.

Timing

- 95. A **T-state** equals **1** clock period of the microprocessor.
- 96. An opcode-fetch machine cycle takes **4** T-states (up to **6** for some instructions).
- 97. An 8085 instruction takes **1 to 5** machine cycles to complete.
- 98. 8085 instructions range from **4** T-states (shortest) to **18** T-states (CALL, longest).
- 99. **Memory read** and **memory write** machine cycles each take **3** T-states.

Interrupts

- 100. The 8085 has **5** hardware interrupts: **TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR**.
- 101. The interrupt priority order (high → low) is: **TRAP > RST 7.5 > RST 6.5 > RST 5.5 > INTR**.
- 102. There is exactly **1** non-maskable 8085 interrupt — **TRAP** — and it has the **highest** priority.
- 103. There is exactly **1** non-vectored 8085 interrupt — **INTR** — which has the **lowest** priority.
- 104. **TRAP** uses vector address **0024H** and is triggered by both **edge and level**.
- 105. **RST 7.5** uses vector address **003CH** and is **rising-edge** triggered.
- 106. **RST 6.5** uses vector address **0034H** and is **level** triggered.
- 107. **RST 5.5** uses vector address **002CH** and is **level** triggered.
- 108. The vector address of any RST n is found by **1** formula — **$n \times 8$** — so **RST 6.5** → $6.5 \times 8 = 0034H$.
- 109. **TRAP** is alternatively called **RST 4.5**, since $4.5 \times 8 = 36 = 0024H$.
- 110. The 8085 has **8** software interrupts: **RST 0 through RST 7**.
- 111. Software interrupt vector addresses run from **0000H (RST 0)** to **0038H (RST 7)** in steps of **8**.
- 112. **RST 5** vectors to **0028H**, while **RST 7** vectors to **0038H**.
- 113. Interrupts are globally controlled by **2** instructions: **EI** (enable) and **DI** (disable).
- 114. The 3 RST interrupts are individually masked by **1** instruction — **SIM** (Set Interrupt Mask).
- 115. Interrupt mask/pending status is read by **1** instruction — **RIM** (Read Interrupt Mask).

116. Interrupts are classified **3** ways: hardware/software, maskable/non-maskable, and vectored/non-vectored.
117. When an interrupt is accepted, the 8085 saves **1** item on the stack — the **Program Counter (PC)**.

8259 Programmable Interrupt Controller

118. A single **8259 PIC** manages **8** interrupt request lines: **IR0–IR7**.
119. In fully-nested mode the 8259 assigns highest priority to **IR0** and lowest to **IR7**.
120. Cascading **1** master with **8** slave 8259s gives up to **64** interrupt lines.
121. The 8259 is initialised by **4** command words: **ICW1, ICW2, ICW3, and ICW4**.
122. The 8259 is operated by **3** command words: **OCW1, OCW2, and OCW3**.
123. The 8259 supports modes like **fully nested, rotating priority, special mask, and polled**.

Memory Classification

124. Memory is broadly classified into **2** types: **primary (main)** and **secondary (mass)** memory.
125. Primary memory examples are **3**: **RAM, ROM, and cache**.
126. Secondary memory examples include **HDD, SSD, CD/DVD, and pen drives**.
127. Memory is also split into **2** classes by power retention: **volatile** and **non-volatile**.
128. Volatile memory loses data on power-off and includes **RAM (SRAM and DRAM)**.
129. Non-volatile memory retains data and includes **ROM, Flash, HDD, and SSD**.
130. The **fastest** but smallest primary memory is **cache**, sitting between the CPU and main memory.

RAM and ROM

131. RAM has **2** main types: **SRAM** and **DRAM**.
132. **SRAM** stores **1** bit using a **flip-flop** (about **6** transistors) and needs **no** refresh.
133. **DRAM** stores **1** bit using **1 transistor + 1 capacitor** and **must** be refreshed periodically.
134. Between the two, **SRAM** is **faster** and costlier; **DRAM** is **denser** and cheaper.
135. The typical use of **SRAM** is **cache**, while **DRAM** is used as **main memory**.
136. ROM has **5** common types: **Mask ROM, PROM, EPROM, EEPROM, and Flash**.
137. **PROM** can be programmed **1** time only by the user (one-time programmable).
138. **EPROM** is erased by **1** method — **ultraviolet (UV) light** — through a quartz window.
139. **EEPROM** is erased **electrically** and can be reprogrammed **byte-by-byte**.
140. **Flash** memory is a block-erasable form of EEPROM used in **USB drives and SSDs**.
141. A memory chip of $2^n \times m$ size has **n** address lines and **m** data lines, so **1K × 8** has **10** address lines.
142. The 8085 distinguishes memory from I/O using **1** signal — **IO/M** — where memory access means **IO/M = 0**.

Flip-Flops

143. There are **4** basic flip-flop types: **SR, JK, D, and T**.
144. A latch differs from a flip-flop on **1** point: latches are **level-triggered**, flip-flops are **edge-triggered**.
145. The **SR** flip-flop has **1** forbidden input combination — **S = R = 1**.
146. The **JK** flip-flop fixes the SR invalid state by **toggling** when **J = K = 1**.
147. The **D** flip-flop has **1** input and its output simply **follows** D ($Q_{\text{new}} = D$).
148. The **T** flip-flop toggles when **T = 1** and is mainly used to build **counters**.
149. The **race-around** condition occurs in **1** flip-flop — the **JK** — when **J = K = 1**.
150. Race-around is removed by **1** technique — the **master-slave** configuration (or edge-triggering).
151. The characteristic equation of the JK flip-flop is $Q_{\text{new}} = JQ' + K'Q$.
152. The **D** flip-flop is preferred for **registers** because it has **no** invalid or race-around state.

Counters

153. A counter with **n** flip-flops has a maximum modulus of 2^n and counts **0 to $2^n - 1$** .

- 154. Counters are **2** types by clocking: **asynchronous (ripple)** and **synchronous**.
- 155. In an **asynchronous** counter the flip-flops are clocked **one after another** (slower).
- 156. In a **synchronous** counter **all** flip-flops share **1** common clock (faster).
- 157. By direction, counters are **3** kinds: **up**, **down**, and **up/down**.
- 158. A **ring counter** with **n** flip-flops produces **n** states (**MOD-n**).
- 159. A **Johnson counter** with **n** flip-flops produces **2n** states (**MOD-2n**).
- 160. The ring counter circulates **1** logic '1' around the register.
- 161. A **MOD-N** counter needs $n = \lceil \log_2 N \rceil$ flip-flops, so a **MOD-10** decade counter needs **4** flip-flops.
- 162. A **MOD-16** counter requires exactly **4** flip-flops, since $2^4 = 16$.

Shift Registers

- 163. Shift registers come in **4** types: **SISO**, **SIPO**, **PISO**, and **PIPO**.
- 164. **SIPO** acts as a **serial-to-parallel** converter, while **PISO** acts as **parallel-to-serial**.
- 165. An **n-bit** register uses **n** flip-flops, and an **n-bit SISO** needs **n** clock pulses to load fully.

Multiplexer and Demultiplexer

- 166. A **MUX** has 2^n inputs, **n** select lines, and **1** output (many-to-one).
- 167. A **DEMUX** has **1** input, **n** select lines, and 2^n outputs (one-to-many).
- 168. A MUX is also called a **data selector**; a DEMUX is also called a **data distributor**.
- 169. An **8:1 MUX** needs **3** select lines, and a **16:1 MUX** needs **4** select lines.
- 170. The number of select lines equals $\log_2(\text{data lines})$, so a **4:1 MUX** uses **2** select lines.

Encoder and Decoder

- 171. An **encoder** has 2^n inputs and **n** outputs (compresses data).
- 172. A **decoder** has **n** inputs and 2^n outputs (expands data).
- 173. A common decoder is the **3-to-8** (binary-to-octal); a common encoder is the **8-to-3** (octal-to-binary).
- 174. A **priority encoder** resolves multiple active inputs by selecting the **highest-priority** one.
- 175. A **BCD-to-7-segment** decoder drives **7** display segments from a **4-bit** BCD input.

Comparator and Parity

- 176. A digital comparator produces **3** outputs: **A > B**, **A = B**, and **A < B**.
- 177. The 1-bit equality output of a comparator is given by **1** gate — **XNOR** ($A = B$ when $A \text{ XNOR } B = 1$).
- 178. Parity is of **2** types: **even parity** and **odd parity**.
- 179. **Even parity** keeps the total number of 1s **even**; **odd parity** keeps it **odd**.
- 180. Even-parity generators use **XOR** gates, while odd-parity generators use **XNOR** gates.
- 181. A parity bit detects an **odd** number of bit errors but can detect **0** even-error cases and **cannot** correct them.

Adders and Subtractors

- 182. A **half adder** has **2** inputs and **2** outputs, with **Sum** = $A \oplus B$ and **Carry** = $A \cdot B$.
- 183. A **full adder** has **3** inputs (A, B, Cin) and **2** outputs, with **Sum** = $A \oplus B \oplus \text{Cin}$.
- 184. A full adder can be built from **2** half adders plus **1** OR gate.
- 185. A **half subtractor** gives **Difference** = $A \oplus B$ and **Borrow** = $A' \cdot B$.
- 186. A **full subtractor** has **3** inputs and gives **Difference** = $A \oplus B \oplus \text{Bin}$.
- 187. An **n-bit** ripple-carry adder is made by chaining **n** full adders.

DAC (Digital-to-Analog)

- 188. There are **2** main DAC types: the **binary-weighted resistor** DAC and the **R-2R ladder** DAC.
- 189. The **binary-weighted** DAC needs many precise resistors (R, 2R, 4R, 8R...), which is its main **disadvantage**.

- 190. The **R-2R ladder** DAC uses only **2** resistor values — **R and 2R** — making it the most popular.
- 191. An **n-bit** DAC produces **2^n** discrete output levels.
- 192. DAC resolution (step size) is given by **$V_{ref} / (2^n - 1)$** for an n-bit converter.
- 193. The R-2R ladder DAC is favoured for **2** reasons: easy **fabrication** and high **accuracy/scalability**.

ADC (Analog-to-Digital)

- 194. Common ADC types are **4**: **flash, successive-approximation (SAR), counter, and dual-slope**.
- 195. The **fastest** ADC is the **flash (parallel)** type, but it is the most **expensive**.
- 196. An **n-bit flash ADC** needs **$2^n - 1$** comparators.
- 197. The **SAR** ADC uses a **binary-search** method and takes **n** clock cycles for **n** bits.
- 198. The **dual-slope** ADC is the **slowest** but most **accurate**, used in digital multimeters.
- 199. ADC performance is judged by **3** parameters: **resolution, accuracy, and conversion time**.
- 200. **Conversion time** is lowest for the **flash** ADC and highest for the **dual-slope** ADC.

Computer Hardware Basics

- 201. The traditional chipset is split into **2** parts: the **Northbridge** (CPU/RAM/graphics) and the **Southbridge** (I/O).
- 202. The **8255 PPI** has **3** ports — **A, B, and C** — providing **24** programmable I/O lines.
- 203. The 8255 supports **3** operating modes: **Mode 0** (simple I/O), **Mode 1** (handshake), and **Mode 2** (bidirectional).
- 204. Port C of the 8255 can be split into **2** nibbles: **upper (PC7–PC4)** and **lower (PC3–PC0)**.
- 205. The **8253/8254** timer contains **3** independent **16-bit** counters and supports **6** modes (Mode 0–5).
- 206. The **8254** is the enhanced **8253**, supporting higher clock frequencies up to about **10 MHz**.
- 207. A PC **SMPS** outputs **3** main DC rails: **+3.3 V, +5 V, and +12 V**.
- 208. Common expansion slots are **2** generations: **PCI** and **PCIe** (PCI Express).